Exp No:6. Experiment to understand feature scaling

Code:

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler

df = pd.DataFrame({
   'Country': ['USA', 'France', 'USA', 'China', 'France', 'USA', 'China', 'France', 'USA', 'China'],
   'Age': [22, 34, 27, np.nan, 45, 38, 31, 50, 29, 41],
   'Salary': [35000, 60000, 48000, 72000, np.nan, 55000, 68000, 80000, 52000, 64000],
   'purchased': ['NO', 'Yes', 'NO', 'Yes', 'NO', 'Yes', 'NO', 'Yes', 'NO', 'Yes']
})

print("Original Dataset.")
print(df)

print("\n\nDataFrame Information")
print(df.info())
```

```
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

age = SimpleImputer(strategy="mean", missing_values=np.nan)
Salary = SimpleImputer(strategy="mean", missing_values=np.nan)

age.fit(features[:, [1]])
Salary.fit(features[:, [2]])

features[:, [1]] = age.transform(features[:, [1]])
features[:, [2]] = Salary.transform(features[:, [2]])

print("\n\nAfter Filling Missing Values.")
print(features)

oh = OneHotEncoder(sparse_output=False)
country = oh.fit_transform(features[:, [0]])

print("\n\nOneHot Encoding:")
print(country)

final_set = np.concatenate((country, features[:, [1, 2]]), axis=1)
```

```
print("\n\nFinal set:")
print(final_set)

sc = StandardScaler()
sc.fit(final_set)
feat_standard_scaler = sc.transform(final_set)

print("\n\nStandard Scaler:")
print(feat_standard_scaler)

mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)
feat_minmax_scaler = mms.transform(final_set)

print("\n\nMin Max Scaler:")
print(feat_minmax_scaler)
```

Output:

```
Original Dataset:
   Country    Age    Salary Purchased
0      USA  22.0  35000.0         No
1   France  34.0  60000.0        Yes
2      USA  27.0  48000.0         No
3    China   NaN  72000.0        Yes
4   France  45.0      NaN         No
5      USA  38.0  55000.0        Yes
6    China  31.0  68000.0         No
7      NaN  50.0  80000.0        Yes
8   France  29.0  52000.0         No
9      USA  41.0  64000.0        Yes


DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #    Column      Non-Null Count  Dtype
---   ------      --------------  -----
 0    Country     9 non-null      str
 1    Age         9 non-null      float64
 2    Salary      9 non-null      float64
 3    Purchased   10 non-null     str
dtypes: float64(2), str(2)
memory usage: 312.0 bytes
None
After Filling Missing Values:
[['USA' 22.0 35000.0]
 ['France' 34.0 60000.0]
 ['USA' 27.0 48000.0]
 ['China' 35.22222222222222 72000.0]
 ['France' 45.0 59333.333333333336]
 ['USA' 38.0 55000.0]
 ['China' 31.0 68000.0]
 ['USA' 50.0 80000.0]
 ['France' 29.0 52000.0]
 ['USA' 41.0 64000.0]]


One Hot Encoding:
[[0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
Final Set:
[[0.0 0.0 1.0 22.0 35000.0]
 [0.0 1.0 0.0 34.0 60000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [1.0 0.0 0.0 35.22222222222222 72000.0]
 [0.0 1.0 0.0 45.0 59333.333333333336]
 [0.0 0.0 1.0 38.0 55000.0]
 [1.0 0.0 0.0 31.0 68000.0]
 [0.0 0.0 1.0 50.0 80000.0]
 [0.0 1.0 0.0 29.0 52000.0]
 [0.0 0.0 1.0 41.0 64000.0]]


Standard Scaler:
[[-5.00000000e-01 -6.54653671e-01  1.00000000e+00 -1.63305082e+00
  -2.00154053e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00 -1.50954278e-01
   5.48367268e-02]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00 -1.01551060e+00
  -9.32224356e-01]
 [ 2.00000000e+00 -6.54653671e-01 -1.00000000e+00  0.00000000e+00
   1.04189781e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00  1.20763422e+00
   5.98484550e-16]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  3.43077904e-01
  -3.56438724e-01]
 [ 2.00000000e+00 -6.54653671e-01 -1.00000000e+00 -5.21478414e-01
   7.12877448e-01]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  1.82517445e+00
   1.69993853e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00 -7.68494505e-01
  -6.03203995e-01]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  7.13602040e-01
   3.83857088e-01]]
```

Result:

Hence, the importance of feature scaling is understood.